Programming and Spreadsheet Modeling

Mathematical Model Building

$f(x)$

$f(x)$

Management Science Models
Mathematical Model Elements

- Assumed values:
  - Parameters
- Dependent variables:
  - Takes on the value of what we're trying to model
- Independent variables:
  - Takes on the values of those things which impact the value of the independent variable
- Mathematical relationship:
  - Function relating the parameters, independent and dependent variables

$f(x)$

Mathematical Models
Business Decision: Break-even volume

- A company must pay a fixed cost of $5,000 to setup their production business.

- Once the business is setup the company will:
  - Pay a cost of $55 to produce each unit to sell
  - Sell each unit for $75

- Before it decides to start the business, management wants to know:
  - How many units must be sold to break even, where total revenues equal total costs and profit will equal zero?
  - How much money will be required to reach break even, covering all costs, including both setup and production?

Management Science Models
Model Categorizations

- Descriptive:
  - *I know the function, but I don't know what Y will be for different Xs.*
  - ***Describes*** *what happens to Y for values of X*

- Predictive:
  - *I know past Xs and Ys, but I don't know the function connecting X to Y.*
  - *Finding the relationship enables* ***predicting*** *Y for any X*

- Prescriptive:
  - *I know the function, and I choose X to get the Y that I want.*
  - ***Prescribes*** *the values of X to obtain the desired value of Y*

$f(x)$

Management Science Models
Break Even Volume Model - Descriptive

- **Situation**
  - A start-up wants to determine the number of units that must be sold to break even.
- **Independent variable values (X)**: **known or assumed**
  - Selling price(p), fixed setup cost (c_f), variable production cost (c_v)
- **Dependent variable values (Y):** **what we want to know**
  - Number of units (V_{BE}) and total cost (TC) until breaking even
- **Function / relationship:** **known**
  - $V_{BE} = {}^{c_f}/_{(p - c_v)}$
- **What is known?**
  - We know the mathematical relationship between price, costs and volume.
- **Model Purpose**
  - Describe what volume ***Y*** will bring us to breakeven given the assumed values for ***X***

Break-even Calculation

Volume that gives a profit of $0

$f(x)$

Break-even Calculation
Parameters and variables

Parameters:

- Assumed or uncontrollable inputs
- Known values

Variables:

- Controllable inputs
- Values to determine

Parameters for this problem:

- Fixed cost to setup the business = \$5,000
- Variable cost to produce each unit = \$55
- Price each unit is sold for = \$75

Variables for this problem:

- V = volume: # of units produced
- V_{BE} = break-even volume:
  # of units produced and sold such that total revenue equals total cost
- Z = total profit (in \$)
- TR = total revenue (in \$)
- TC = total cost (in \$)

Assign letters for the parameters:

- Fixed cost c_f = \$5,000
- Variable cost c_v = \$55
- Selling price p = \$75

Break-even Calculation
Revenue and cost formula derivations

Total Cost (TC): amount of money spent to setup production and produce all units

- $c_f = \$5{,}000$ (fixed cost to setup the business)
- $c_v = \$55$ (variable cost to produce a single unit)
- V = volume (volume = the number of units produced)
- $TC(V) = c_f + c_v V$ (function that relates the units produced to total cost)
- $\boldsymbol{TC(V) = \$5{,}000 + \$55V}$

Total Revenue (TR): amount of money received for selling the units

- $p = \$75$ (selling price per unit)
- $TR(V) = pV$ (function that relates the revenue to units sold)
- $\boldsymbol{TR(V) = \$75V}$

When should you switch parameters from letters to numeric values?

Break-even Calculation
Profit and Break-even volume formula derivations

Profit (Z): how much money is retained after producing and selling the units

- Profit = Total Revenue – Total Cost

  Profit is a function of the revenue and cost

- $Z(V) = TR(V) - TC(V)$

  $Z(V) = pV - (c_f + c_v V)$

  Higher level function notation shows the big picture of what's happening, while expanding them enables solving for specific values

Break-even volume (V_{BE}) is that volume which gives a profit of $0

- $Z(V) = TR(V) - TC(V)$

  $Z(V_{BE}) = 0$

  V_{BE} is **a specific value of** V where profit $Z(V_{BE}) = 0$

  $0 = p\,V_{BE} - (c_f + c_v\,V_{BE})$

  replace TR and TC with their formulas

  $p\,V_{BE} = (c_f + c_v\,V_{BE})$

  TR must equal TC to give profit $Z(V_{BE}) = 0$

  $p\,V_{BE} - c_v\,V_{BE} = c_f$

  subtract $c_v V_{BE}$ from both sides, to get V_{BE} on one side

  $(p - c_v)V_{BE} = c_f$

  collect terms where $(p - c_v)$ = marginal profit

  $$\boldsymbol{V_{BE} = \frac{c_f}{(p - c_v)}}$$

  solve for V_{BE} by dividing both sides by $(p - c_v)$

Break-even Calculation
Break-even problem solution

Break-even volume (V_{BE}) is that volume which gives a profit $Z(V_{BE})$ of $0

$Z(V) = TR(V) - TC(V)$

$Z(V_{BE}) = 0$

$0 = p\,V_{BE} - (\,c_f + c_v\,V_{BE}\,)$ replace TR and TC with their formulas

$0 = 75\,V_{BE} - (\,5{,}000 + 55\,V_{BE}\,)$ replace parameters p, c_f, c_v with their values

$75\,V_{BE} = (5{,}000 + 55\,V_{BE}\,)$ TR must equal TC to give profit $Z(V_{BE}) = 0$

$75\,V_{BE} - 55\,V_{BE} = 5{,}000$ subtract $c_v V_{BE}$ from both sides

$(\,75 - 55\,)V_{BE} = 5{,}000$ collect terms where $(p - c_v)$ = marginal profit

$V_{BE} = \dfrac{5{,}000}{(\,75 - 55)}$ isolate V_{BE} by dividing both sides by $(75 - 55)$

$\boldsymbol{V_{BE} = 250\ units}$

Once the company has sold 250 units, the fixed cost of 5,000 and all variable costs to that point will have been paid off.

Break-even Calculation
Break-even problem – the next question

At the break-even volume V_{BE} how much has the firm paid out in costs?

$V_{BE} = 250\ units$

The company has sold 250 units and using the profits from selling those 250 units has paid off the fixed cost of 5,000 to setup the business.

But how much did it cost to setup the business AND produce the units to sell?

- $TC(V) = c_f + c_v V$ function that relates the units produced to total cost
- $TC(V) = \$5{,}000 + \$55V$ the general total cost function for our company
- $TC(V_{BE}) = \$5{,}000 + \$55V_{BE}$ the specific total cost function at break-even point
- $TC(250) = \$5{,}000 + \$55(250) = \$5{,}000 + \$13{,}750$
- $TC(V_{BE}) = \mathbf{\$18{,}750}$ the company needs $18,750 before it will start making money

Break-even Calculation
Directional analysis – back of the envelope

What is going on? The firm makes a marginal profit $(p - c_v)$ on each item it sells and uses that marginal profit to pay off it's fixed cost c_f.

- $V_{BE} = \frac{c_f}{(p - c_v)}$ V_{BE} is the fixed cost c_f divided by the marginal profit $(p - c_v)$

How does the break-even volume (V_{BE}) change if the parameters change?

- $\uparrow V_{BE} = \frac{\uparrow c_f}{(p - c_v)}$ Will V_{BE} increase or decrease if the fixed cost c_f increases?
- $\downarrow V_{BE} = \frac{c_f}{\uparrow (p - c_v)}$ Will V_{BE} increase or decrease if the marginal profit $(p - c_v)$ increases?
- $\uparrow V_{BE} = \frac{c_f}{\downarrow (p - c_v)}$ Will V_{BE} increase or decrease if the selling price p decreases?